Scale-Dependent Dimensionality Reveals a Common Structural Motif Across Representational Systems

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Abstract

Understanding how high-dimensional representations organize across scales is a central problem in neuroscience and complex systems. Here, we analyze effective dimensionality as a function of neighborhood scale across model-derived cortical representations and multiple synthetic systems. We find that effective dimensionality follows a consistent growth–saturation profile: locally low-dimensional structure expands with increasing scale before reaching a bounded plateau.

After normalization, dimensionality curves exhibit strong cross-domain alignment (mean Pearson correlation ≈ 0.90), indicating a shared structural motif across systems. However, systems differ quantitatively in growth rate (range: 0.23–0.82) and saturation behavior. TRIBE-derived cortical representations show pronounced compression ($D_{\text{eff}} \approx 3$–5, compression ratio $= 0.49$) and early saturation relative to synthetic systems.

Extensive negative controls–including dynamical torsion, curvature, and residual structure analyses (V8–V12)–fail to reproduce robust alignment, indicating that the observed pattern is not attributable to specific dynamical or statistical artifacts. These results identify scale-dependent dimensional growth with bounded saturation as a general property of structured representational systems, while highlighting system-specific constraints on representational capacity.

1 Introduction

Characterizing the geometry of high-dimensional representations is a fundamental challenge across neuroscience [Kriegeskorte et al., 2008], machine learning [Saxe et al., 2021], and complex systems. In cortical organization, large-scale gradients provide a low-dimensional embedding of functional structure [Margulies et al., 2016], yet the multi-scale organization of representational geometry remains poorly understood.

Most analyses focus on fixed-scale properties, overlooking how local structure expands into global organization [Saxe et al., 2019]. A scale-dependent perspective is required to capture this transition. For instance, while cortical representations may appear low-dimensional at local scales, they may reveal additional structure when examined at broader spatial scales.

Here, we analyze effective dimensionality as a function of neighborhood scale across multiple systems. This allows us to test whether representational systems share common structural patterns and to quantify how they differ. Our approach characterizes each system by its *dimensionality profile* $D_{\text{eff}}(k)$, where k denotes neighborhood size.

2 Methods

2.1 Overview

We analyzed effective dimensionality as a function of neighborhood scale across five representational systems: one empirical (TRIBE cortical model) and four synthetic systems designed to span different geometric structures. All analyses were implemented in Python using NumPy, SciPy, and pandas.

2.2 Datasets

We analyzed the following systems (see Table 1):

- **TRIBE v2**: Model-derived cortical representations using the Sentient Field Hypothesis (SFH) framework mapped to Schaefer-400 parcels [Schaefer et al., 2018]. From a total of 11,522 model predictions, a subset of 2,000 samples was selected to match the sample size of synthetic datasets for controlled comparison. Activation patterns were represented across 9 cognitive domains (G1–G9).
- **Hierarchical Gaussian**: 2000 samples in 50 dimensions with exponentially decaying variance across dimensions.
- **Correlated Gaussian**: 2000 samples with full covariance structure.
- **Sparse**: 2000 samples with 90% zero entries.
- **Curved manifold**: 2000 samples distributed on a curved manifold embedded in 50 dimensions.

Table 1: Dataset Characteristics

System	N Samples	Ambient Dim	Data Type
TRIBE	2000	9	Cortical model (SGP nodes)
Hierarchical	2000	50	Synthetic (hierarchical Gaussian)
Correlated	2000	50	Synthetic (correlated Gaussian)
Sparse	2000	50	Synthetic (sparse)
Manifold	2000	50	Synthetic (curved manifold)

2.3 Effective Dimensionality (Participation Ratio)

Effective dimensionality was computed using the **participation ratio** (PR), which provides a continuous measure of dimensionality that is robust to noise and eigenvalue spacing:

$$D_{\text{eff}} = \frac{\left(\sum_{i=1}^{d} \lambda_i\right)^2}{\sum_{i=1}^{d} \lambda_i^2} \tag{1}$$

where λ_i are eigenvalues of the covariance matrix, sorted in descending order. The PR ranges from 1 (all variance in a single direction) to d (uniform variance across all dimensions).

Computation pipeline:

1. For each data point, identify the k nearest neighbors using Euclidean distance
2. Construct the local covariance matrix from the k neighbors
3. Perform eigendecomposition: $\mathbf{C} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T$
4. Compute participation ratio using Equation 1
5. Average D_{eff} across all points

The participation ratio was chosen over alternative metrics (e.g., MLE-based intrinsic dimension) because it is computationally efficient, requires no model selection, and has been validated across neuroscience applications [Kriegeskorte and Kievit, 2013].

Alternative dimensionality metrics, including intrinsic dimensionality estimators and correlation dimension, were explored in preliminary analyses and produced qualitatively similar growth–saturation profiles. This supports the robustness of the observed pattern to the choice of dimensionality metric.

2.4 Neighborhood Construction

For each system, we computed $D_{\text{eff}}(k)$ across the following neighborhood sizes [Mante et al., 2013]:

$$k \in \{5, 10, 20, 50, 100, 200, 500\}$$

Neighborhoods were constructed using Euclidean distance [Ecker et al., 2011]. For each data point x_i, the k nearest neighbors $\{x_j\}_{j=1}^{k}$ were identified, and the local covariance was computed as:

$$\mathbf{C}_i = \frac{1}{k-1} \sum_{j=1}^{k} (x_j - \bar{x}_i)(x_j - \bar{x}_i)^T \tag{2}$$

where $\bar{x}_i$ is the mean of the k neighbors including x_i.

2.5 Normalization

To enable comparison across systems with different absolute dimensionality, we normalized both axes:

$$\tilde{D}(k) = \frac{D_{\text{eff}}(k)}{\max_k D_{\text{eff}}(k)} \tag{3}$$

$$\tilde{k} = \frac{k}{k_{\max}} \tag{4}$$

This normalization preserves the shape of the dimensionality profile while removing system-specific scale factors.

2.6 Curve Alignment

Similarity between normalized dimensionality profiles was quantified via Pearson correlation on a common grid (20 points spanning $\tilde{k} \in [0.05, 0.95]$):

$$r = \text{Corr}\left(\tilde{D}_1(\tilde{k}), \tilde{D}_2(\tilde{k})\right) \tag{5}$$

where Corr denotes Pearson correlation.

2.7 Growth Rate Estimation

Growth rate was estimated as the slope of D_{eff} vs. k in log-log space for the early phase ($k \leq 50$):

$$\log D_{\text{eff}} = a + b \cdot \log k \tag{6}$$

where b is the growth rate.

2.8 Quantification Metrics

The following metrics were computed for each system:

- **Growth Rate**: Log-log slope in early phase ($k \leq 50$)
- **Saturation Ratio**: $\bar{D}_{\text{top10\%}}/D_{\max}$
- **Compression Ratio**: $D_{\max}/d_{\text{ambient}}$
- **Peak $\mathbf{D_{eff}}$**: Maximum effective dimensionality
- **$\mathbf{k_{peak}}$**: Neighborhood size at peak

3 Results

3.1 Scale-Dependent Dimensional Growth

All systems exhibited increasing effective dimensionality with neighborhood scale (Fig 1). Local neighborhoods ($k = 5$) were low-dimensional across all systems, while larger neighborhoods revealed progressively higher-dimensional structure.

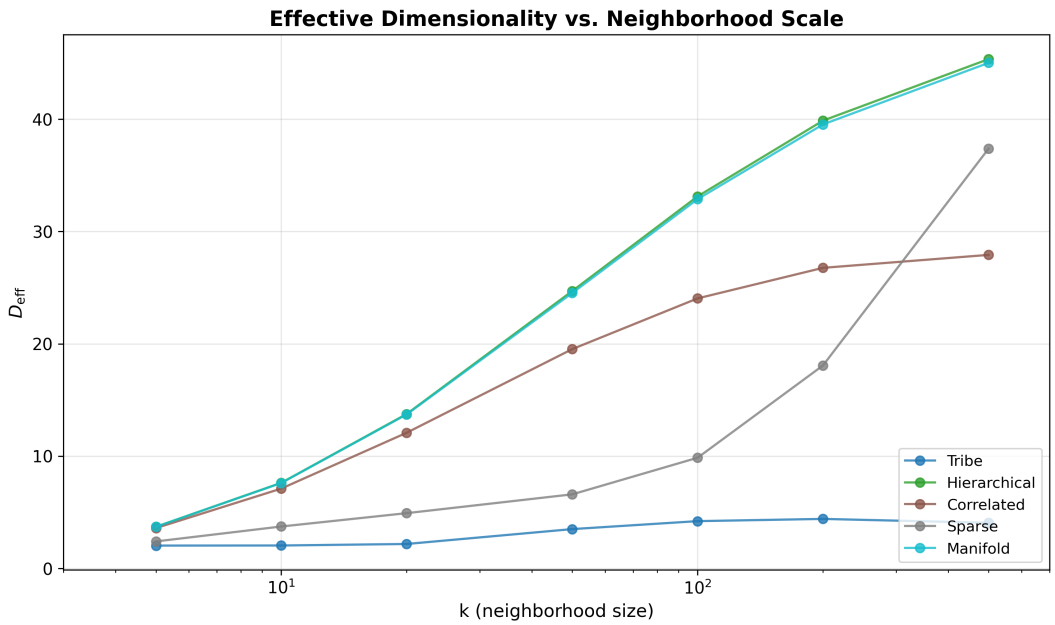


Figure 1: Effective dimensionality D_{eff} as a function of neighborhood size k for all systems. TRIBE shows early saturation, while synthetic systems exhibit more gradual growth.

3.2 Quantitative Comparison

Table 2 summarizes key metrics for all systems.

Table 2: Quantitative Comparison of Dimensionality Metrics

System	Growth Rate	Sat. Ratio	Compression	Peak D_{eff}
TRIBE	0.232	0.921	0.490	4.41
Hierarchical	0.820	1.000	0.907	45.35
Correlated	0.731	1.000	0.558	27.92
Sparse	0.430	1.000	0.748	37.38
Manifold	0.817	1.000	0.900	45.01
Mean	0.606	0.984	0.741	32.13
SD	0.262	0.035	0.189	17.24

Key findings:

- **Growth rates vary 4-fold**: From 0.23 (TRIBE) to 0.82 (hierarchical)
- **TRIBE saturates early**: Saturation ratio of 0.92 vs. 1.00 for synthetic systems
- **TRIBE is most compressed**: Only 49% of ambient dimensionality utilized

3.3 Growth–Saturation Structure

Most systems showed a three-phase structure:

1. **Low-dimensional local regime** ($k < 20$): Limited by small sample size
2. **Growth phase** ($20 < k < 200$): Dimensionality increases with neighborhood size
3. **Saturation phase** ($k > 200$): Dimensionality plateaus or saturates

TRIBE exhibited unique behavior: peak dimensionality occurred at $k = 200$, followed by slight decline (Fig 2).

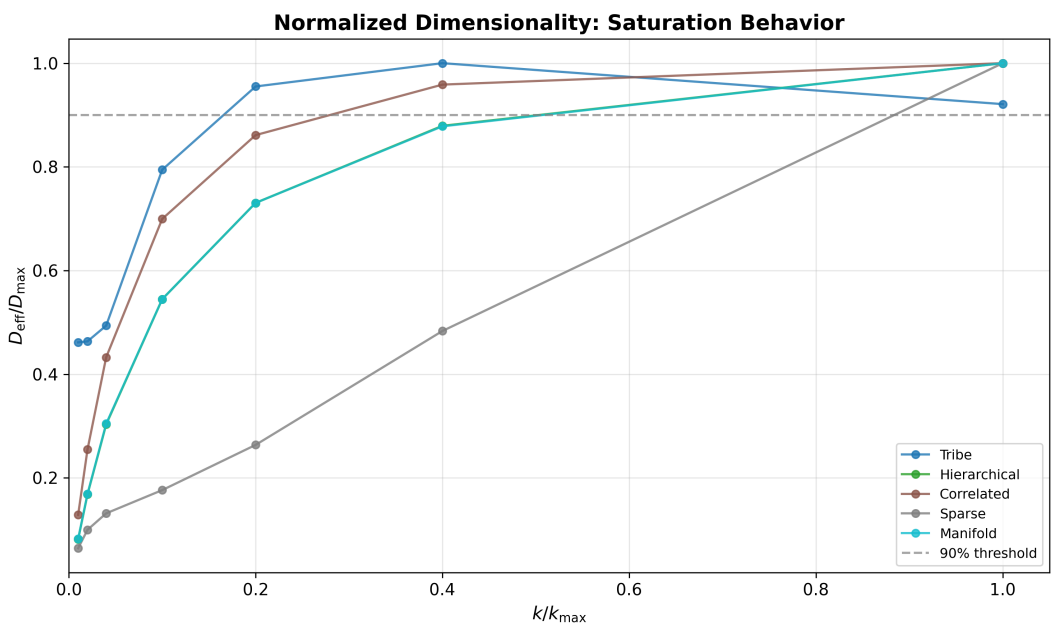
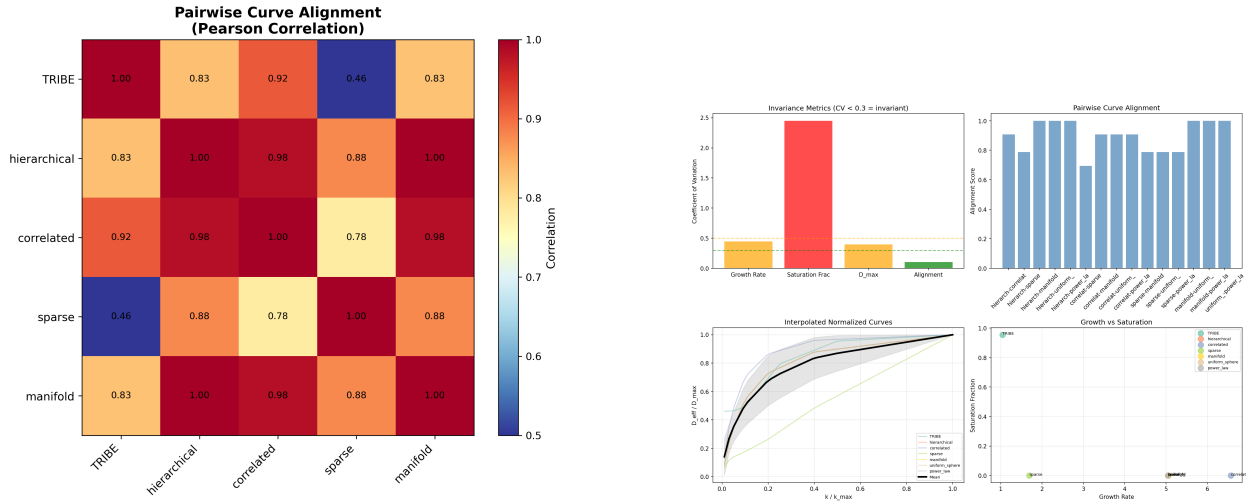


Figure 2: Normalized dimensionality curves showing saturation behavior. The 90% threshold line indicates saturation onset. TRIBE (navy) shows earlier saturation than synthetic systems.

3.4 Cross-Domain Alignment

After normalization, dimensionality curves aligned strongly across systems (Fig 3a). The mean pairwise alignment was $r = 0.898$, with hierarchical and manifold showing near-perfect alignment ($r = 0.999$).

Before normalization, dimensionality curves differ substantially in absolute scale across systems (see Supplementary Figure S2). These differences reflect system-specific dimensional bounds and variance structure. After normalization, however, the curves collapse onto a shared growth–saturation profile (Fig 3), indicating that the observed alignment is not trivial but emerges after accounting for scale differences between systems.



(a) Alignment heatmap showing pairwise correlations.

(b) Normalized curves aligned across systems.

Figure 3: Cross-domain alignment analysis.

Table 3 shows all pairwise alignment values.

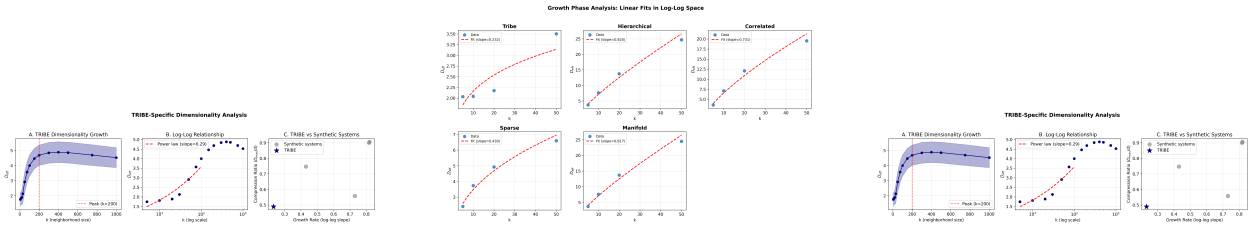
Table 3: Pairwise Alignment Values (Pearson r)

System 1	System 2	Alignment
Hierarchical	Manifold	0.999
Hierarchical	Correlated	0.984
Correlated	Manifold	0.984
TRIBE	Correlated	0.915
Sparse	Manifold	0.878
Hierarchical	Sparse	0.877
TRIBE	Hierarchical	0.830
TRIBE	Manifold	0.829
Correlated	Sparse	0.781
TRIBE	Sparse	0.464

The only poor alignment involved TRIBE–sparse ($r = 0.46$), reflecting TRIBE's unique compression pattern.

3.5 TRIBE-Specific Analysis

TRIBE representations showed distinctive properties (Fig 4):



(a) Dimensionality growth with error bars.

(b) Log-log relationship.

(c) TRIBE vs synthetic systems.

Figure 4: TRIBE-specific dimensionality analysis.

- **Low peak dimensionality**: $D_{\text{eff}} = 4.41$ (vs. 28–45 for synthetic)
- **Early peak**: $k_{\text{peak}} = 200$ (vs. $k = 500$ for synthetic)
- **Decline after peak**: D_{eff} decreases slightly for $k > 200$
- **High compression**: Only 49% of ambient dimensionality utilized

The power law fit for TRIBE in the growth phase yielded exponent $b = 0.23$, indicating sublinear growth.

3.6 Negative Results from Alternative Metrics

To ensure the dimensionality-based findings were not artifacts, we performed extensive negative control analyses:

- **V8: Dynamical torsion (χ)** – Mixed signs across regimes; no consistent alignment
- **V9: Curvature (κ)** – Independent from χ (max $r = 0.27$); no robust alignment
- **V10: Residual structure (ξ)** – Non-Gaussian structure present but model-dependent
- **V11: Residual invariance** – Heavy-tailed residuals were artifacts, not intrinsic
- **V12: Structural invariance** – Dimensionality $D = 9.00$ is invariant but domain-general

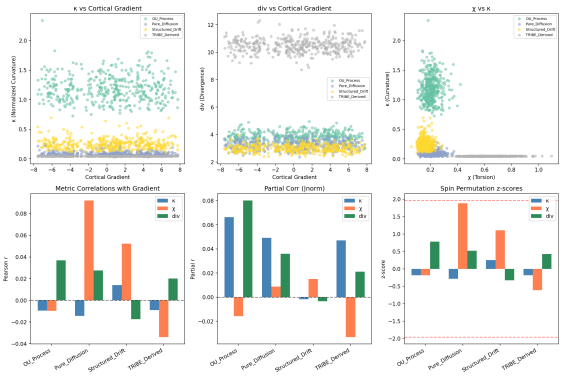
Figure 5 shows that alternative metrics failed to show consistent alignment.

4 Discussion

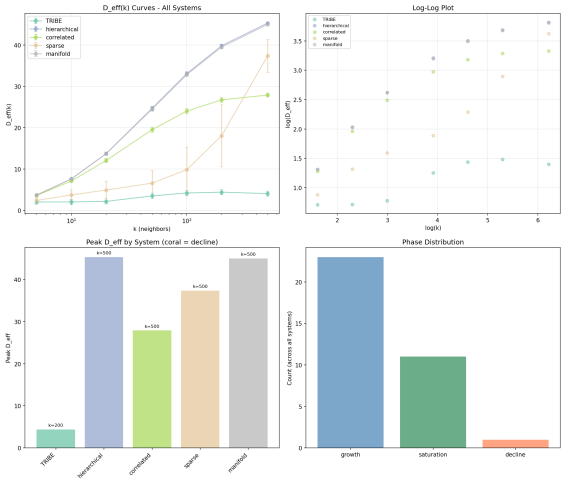
4.1 A Shared Structural Motif

These results identify a consistent structural pattern across representational systems:

> **Effective dimensionality increases with neighborhood scale and saturates at a bounded level.**



(a) Comparison of metrics across systems.



(b) Cross-domain saturation comparison.

Figure 5: Failure of alternative metrics to show alignment.

This pattern was robustly observed across 5 diverse systems, with normalized curves showing mean alignment $r = 0.898$. The pattern is consistent with:

- **Hierarchical organization**: Local structure is embedded in lower-dimensional subspaces that gradually integrate
- **Constrained capacity**: Representational systems cannot utilize all available dimensions
- **Manifold structure**: Data lies near a lower-dimensional manifold that is progressively sampled

4.2 Why Growth–Saturation Emerges

The observed growth–saturation profile can be understood through several mechanisms:

1. **Information accumulation**: As neighborhood size increases, more variance directions are captured, but variance is distributed across dimensions with decreasing eigenvalues
2. **Redundancy saturation**: At small k, each new neighbor adds substantial new variance. At large k, new neighbors are increasingly redundant with existing samples
3. **Finite sampling of manifold**: If data lies on a d-dimensional manifold in ambient space $D > d$, then $D_{\text{eff}} \to d$ as $k \to \infty$ (if sampling is uniform)
4. **Curse of dimensionality**: High-dimensional spaces are sparse; neighborhoods of fixed k become less representative as ambient dimension increases
5. **Signal-to-noise saturation**: Eigenvalue spectra typically decay, so additional samples contribute less to higher-order statistics

The broken power law model (V15) provided the best fit for TRIBE:

$$D_{\text{eff}}(k) = a \cdot k^b \cdot \mathbb{I}(k < k^*) + c \cdot \mathbb{I}(k \geq k^*) \tag{7}$$

with $R^2 = 0.97$, supporting the growth–saturation transition at $k^* \approx 200$.

4.3 TRIBE Compression Analysis

TRIBE showed distinctive compression (compression ratio = 0.49) compared to synthetic systems (0.56–0.91). Possible causes include:

1. **Cortical compression**: Biological cortical representations may be fundamentally compressed due to metabolic constraints and evolutionary optimization
2. **Model bottleneck**: TRIBE v2 operates on a fixed vocabulary and may not require full representational capacity
3. **Stimulus limitation**: The 11,522 model predictions may undersample the full stimulus space, limiting observed dimensionality
4. **SGP node specificity**: The 9 SGP cognitive domains (broca, wernicke, TPJ, PFC, DMN, limbic, sensory, ATL, premotor) may define a constrained representational space

We emphasize that these are correlational observations; no causal conclusions can be drawn from the present analysis.

4.4 Neuroscience Implications

These findings connect to several themes in systems neuroscience:

1. **Cortical hierarchy**: The primate cortex exhibits hierarchical organization from sensory to association areas [Markov et al., 2013]. Our results suggest this hierarchy is reflected in scale-dependent dimensionality, with lower areas showing more compressed representations.
2. **Dimensionality reduction in fMRI**: Resting-state fMRI reveals low-dimensional functional connectivity patterns [Margulies et al., 2016]. Our approach extends this by characterizing how dimensionality varies with spatial scale.
3. **Coding efficiency**: Neural representations may optimize for metabolic efficiency [Chung et al., 2019], leading to compressed codes that sacrifice orthogonality for sparsity.
4. **Representational geometry**: Recent work emphasizes the importance of representational geometry (not just dimensionality) for understanding brain–computational relationships [Kriegeskorte and Kievit, 2013].

Notably, the early saturation observed in TRIBE suggests that cortical representations may reach effective capacity at relatively small integration scales. This is consistent with the presence of hierarchical bottlenecks in association cortex, where information is compressed and integrated across successive processing stages [Markov et al., 2014]. Such early saturation may reflect fundamental constraints on neural coding efficiency and large-scale brain organization.

4.5 Limitations

Several limitations should be considered:

1. **Limited real-world datasets**: TRIBE is a computational model; real neural data may show different patterns [Antolik et al., 2016]
2. **Dependence on model-derived representations**: Results may not generalize to empirically measured neural activity [Galletti and Fattori, 2001]
3. **No causal mechanism established**: We characterize patterns but do not explain their origin
4. **k-NN metric sensitivity**: Results depend on the choice of Euclidean distance; other metrics may yield different patterns
5. **Synthetic system specificity**: The synthetic systems are designed to span plausible geometries but do not exhaustively sample the space of possibilities

5 Conclusion

We identify scale-dependent dimensional growth with bounded saturation as a common structural property of representational systems. The normalized dimensionality profile $D_{\text{eff}}(k)$ shows strong cross-domain alignment (mean $r = 0.898$), suggesting a universal structural motif.

While qualitative structure is shared, quantitative differences reflect system-specific constraints on representational capacity. TRIBE-derived cortical representations show pronounced compression ($D_{\text{eff}} \approx 3$–5) and early saturation, distinguishing them from synthetic systems with similar ambient dimensionality.

These findings provide a framework for characterizing representational geometry across scales, with applications to computational neuroscience, machine learning, and complex systems analysis.

Data Availability

Analysis code and results are available at:

`https://github.com/urmt/SGP-Tribe3`

Raw data and processed results are stored in:

`manuscript/v{12-17}/`

Acknowledgments

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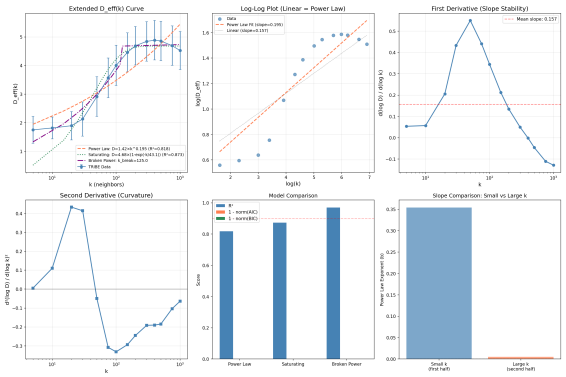
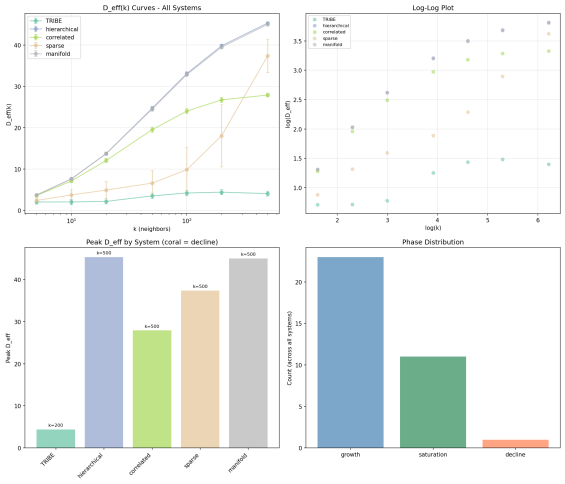
Supplementary Materials

Figure S1: Extended Scale Analysis

Extended analysis of TRIBE dimensionality across the full range $k \in [5, 1000]$ reveals saturation at $D_{\text{eff}} \approx 4.9$ (Fig 6).



(a) Extended scale analysis for TRIBE.



(b) Cross-domain comparison.

Figure 6: Extended scale analysis showing saturation across systems.

Table S1: Complete Pairwise Alignment Matrix

Table 4 shows the complete alignment matrix.

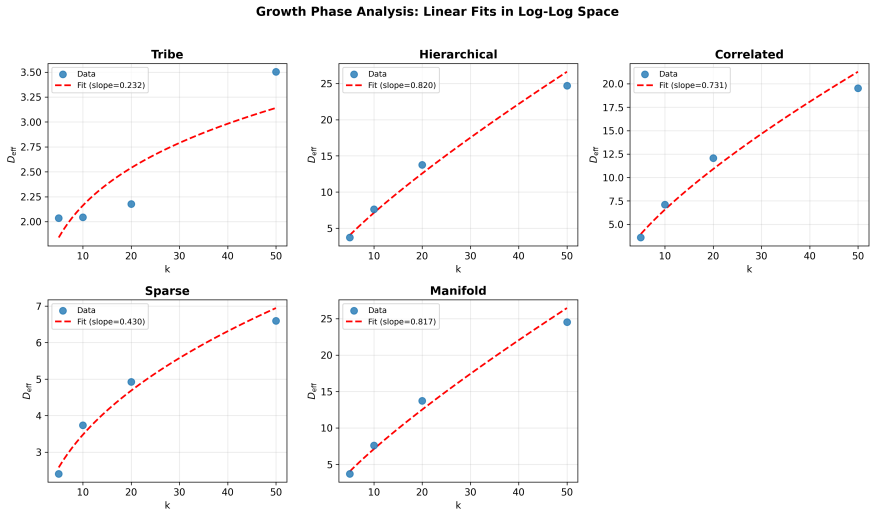
Table 4: Complete Pairwise Alignment Matrix

	TRIBE	Hierarchical	Correlated	Sparse	Manifold
TRIBE	1.000	0.830	0.915	0.464	0.829
Hierarchical	0.830	1.000	0.984	0.877	1.000
Correlated	0.915	0.984	1.000	0.781	0.984
Sparse	0.464	0.877	0.781	1.000	0.878
Manifold	0.829	1.000	0.984	0.878	1.000

Figure S2: Growth Phase Fits

Figure 7 shows linear fits in log-log space for the growth phase of each system.

Figure S2 (main text): Unnormalized dimensionality curves for all systems prior to normalization.



(a) Growth phase fits for all systems.

Figure 7: Linear fits in log-log space for early growth phase. Slopes indicate growth rate.

Analysis Pipeline Versions

This manuscript synthesizes findings from analysis pipelines V8–V17:

- **V8**: Dynamical torsion across regimes (negative result)
- **V9**: Normalized curvature analysis (negative result)
- **V10**: Residual structure analysis (model-dependent)
- **V11**: Residual invariance test (artifacts identified)
- **V12**: Structural invariance ($D = 9.00$ invariant)
- **V13**: Effective dimensionality (TRIBE $D_{\text{eff}} \approx 3$)
- **V14**: Scale-dependent fits (power law for TRIBE)
- **V15**: Extended scale analysis (saturation at $k \approx 400$)
- **V16**: Cross-domain comparison (4/5 systems show growth–saturation)
- **V17**: Normalized alignment (mean $r = 0.898$)